Introduction to Machine Learning

What is Machine Learning?
Machine learning (ML) enables computers to learn patterns from data rather than explicit programming.

Types of Learning
• Supervised Learning
• Unsupervised Learning
• Reinforcement Learning

Training Pipeline
1. Collect data
2. Clean data
3. Split train/test
4. Train model
5. Evaluate results
6. Deploy

Python Example
```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
```

Overfitting
Overfitting occurs when a model memorizes the training data instead of learning general patterns.

Evaluation Metrics
Accuracy, Precision, Recall, F1 Score, RMSE.

Feature Engineering
Normalization, Standardization, One-hot encoding, Feature selection.

Equation
MSE = (1 / n) Σ(y – ■)²

Footnotes
1. Keep train/test independent.
2. Use random seeds.
3. Cross-validation estimates generalization.